CLOUD COMPUTING

College Presentation Report

Topic
Cloud Computing: Concepts, Models, Benefits and Applications

Prepared for
College Seminar / Academic Presentation

Prepared by
Name: _____
Class: _____
Date: _____

Simple English • Clear explanations • Presentation-ready content

1. Introduction to Cloud Computing

Cloud computing is a way of using computing services over the internet. Instead of buying and maintaining all computer hardware and software locally, people and organizations can use resources provided by cloud service companies.

These resources may include storage, servers, databases, networking, software and tools for developing applications. Users can access many cloud services through a web browser or an application, from different locations and devices.

For example, when a student stores a document in Google Drive or watches a video using an online streaming service, cloud-based systems may be working behind the scenes.

2. Meaning and Definition

Definition: Cloud computing is the delivery of computing resources and services over the internet, usually on demand and with flexible usage.

The word **cloud** is a common symbol for the internet. In cloud computing, data and applications can be stored or processed on remote systems in data centres rather than only on a user's personal computer.

Key characteristics

- **On-demand service:** Users can access resources when they need them.
- **Internet access:** Services are commonly available through an internet connection.
- **Shared resources:** A provider can serve many customers using a shared infrastructure with logical separation.
- **Scalability:** Resources can often be increased or reduced as requirements change.
- **Measured usage:** Some services charge according to the resources used.

3. How Cloud Computing Works

Cloud services run on computers and systems in data centres. A user connects to a cloud service through the internet. The provider's systems process the request, store or retrieve information, and send the result back to the user.

- The user opens a website or application and signs in if required.
- The request travels through the internet to the cloud provider.
- Cloud servers process the request or access the required data.
- The result is returned to the user's device.

A stable internet connection is useful for many cloud services. Some applications also provide limited offline features and synchronize changes when the connection returns.

4. Types of Cloud Deployment

Cloud deployment models describe where cloud infrastructure is located and who can use it.

Model	Meaning	Example / Use
Public cloud	Services are provided over the internet to many customers by a cloud provider.	Web hosting, online storage and app services.
Private cloud	Cloud infrastructure is dedicated to one organization.	Organizations with specific control or compliance needs.
Hybrid cloud	Combines private infrastructure with public cloud services.	Keeping some data privately while using public cloud capacity.
Community cloud	Infrastructure is shared by organizations with similar needs.	A group of institutions with common requirements.

5. Cloud Computing Service Models

Service models explain what level of computing service the cloud provider manages and what the customer manages.

Model	Full form and purpose	Examples
IaaS	**Infrastructure as a Service.** Provides basic computing resources such as virtual machines, storage and networks. Customers manage their operating systems and applications.	Amazon EC2, Azure Virtual Machines, Google Compute Engine.
PaaS	**Platform as a Service.** Provides a platform for developers to build, test and deploy applications without managing all underlying infrastructure.	Google App Engine, Azure App Service, Heroku.
SaaS	**Software as a Service.** Provides ready-to-use software over the internet. The provider manages the application and infrastructure.	Google Workspace, Microsoft 365, Salesforce.

Easy way to remember: IaaS provides infrastructure, PaaS provides a development platform, and SaaS provides ready-to-use software.

6. Advantages of Cloud Computing

- **Lower initial cost:** Organizations may avoid buying and maintaining all their own servers.
- **Access from different places:** Users can access services from connected devices and locations.
- **Scalability:** Computing resources can be adjusted as demand changes.
- **Backup and recovery options:** Cloud services can support data backup and disaster recovery when configured correctly.
- **Collaboration:** Multiple users can work on shared files and applications.
- **Maintenance:** The provider manages many parts of the underlying infrastructure.

7. Disadvantages and Challenges

- **Internet dependency:** Some services may be unavailable or limited when the connection is poor.

- **Security and privacy risks:** Data must be protected using suitable access controls, encryption and safe practices.
- **Service outages:** A provider may experience technical problems that temporarily affect access.
- **Ongoing costs:** Usage charges can become high if resources are not monitored.
- **Vendor lock-in:** Moving applications and data from one provider to another may be difficult.
- **Limited control:** Customers may not control every part of the provider's infrastructure.

8. Applications of Cloud Computing

- **Education:** Online classes, digital learning platforms and shared study materials.
- **Business:** Customer management, accounting, collaboration and business applications.
- **Healthcare:** Secure storage and access to suitable digital health information systems.
- **Entertainment:** Music, video streaming and online gaming services.
- **Banking:** Digital banking systems, transaction processing and data services.
- **Software development:** Hosting websites, testing applications and managing development environments.
- **Personal use:** Photo backup, email, online documents and file storage.

9. Popular Cloud Computing Platforms

Platform	Provider	Common services
Amazon Web Services (AWS)	Amazon	Computing, storage, databases, networking and AI services.
Microsoft Azure	Microsoft	Virtual machines, app hosting, databases and analytics.
Google Cloud	Google	Computing, data analytics, storage and machine learning services.
IBM Cloud	IBM	Cloud infrastructure, enterprise and AI-related services.

10. Security in Cloud Computing

Cloud security is a shared responsibility. The cloud provider protects the physical data centres and the services it operates. Customers are responsible for tasks such as managing user access, protecting passwords, choosing secure settings and handling their data appropriately. The exact division depends on the service model.

- Use strong, unique passwords and multi-factor authentication.
- Give users only the access they need.
- Encrypt sensitive data where appropriate.
- Keep backups and test recovery procedures.
- Review permissions, security settings and usage regularly.

11. Future of Cloud Computing

Cloud computing continues to develop as organizations use more digital services. Important trends include hybrid and multi-cloud systems, serverless computing, edge computing, AI and machine learning services, and stronger security tools. These developments can help organizations build flexible services, process data and support new applications.

Cloud computing is likely to remain an important part of education, business, healthcare, entertainment and software development.

12. Conclusion

Cloud computing provides computing resources such as storage, servers, platforms and software through the internet. Its main benefits include flexible access, scalability, collaboration and reduced need for local infrastructure. However, users and organizations must consider security, privacy, internet availability and cost. Understanding cloud deployment and service models helps people choose the right cloud solution for their needs.

13. Short Questions for Presentation / Viva

1. What is cloud computing? It is the delivery of computing services over the internet.

2. What are the three main service models? IaaS, PaaS and SaaS.

3. Name two public cloud providers. Amazon Web Services and Microsoft Azure.

4. What is SaaS? Software as a Service provides ready-to-use software over the internet.

5. Give one advantage of cloud computing. It allows flexible access to computing resources and can scale with demand.

6. Give one challenge of cloud computing. Security and privacy need careful management.

14. References

The following official resources can be used for further reading:

- Amazon Web Services – Cloud Computing: https://aws.amazon.com/what-is-cloud-computing/
- Microsoft Azure – What is cloud computing?: https://azure.microsoft.com/resources/cloud-computing-dictionary/what-is-cloud-computing/
- Google Cloud – Cloud Computing: https://cloud.google.com/learn/what-is-cloud-computing
- IBM – Cloud Computing: https://www.ibm.com/think/topics/cloud-computing

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